LEVEL UP ACADEMY

Class 9 | GSEB / old-book students | Science

Practice Pack: Gravitation, Floatation and Density

Chapter / Topic: Topic-based. Gravitation is reported to be dropped from the new NCERT Class 9 book, so check your syllabus first

Name: _____ Roll No.: _____ Date: _____

Marks obtained: _____

General instructions:

1. Use this pack only if Gravitation is in your students' current syllabus (likely for GSEB / old-book students). It is also good Foundation practice.
2. Key relations: $F = Gm_1m_2/r^2$ · $W = mg$ · $v = u + gt$ · $s = ut + \frac{1}{2}gt^2$ · $v^2 = u^2 + 2gs$ · density = mass/volume. Take $g = 9.8\ m/s^2$ unless told otherwise.

LEVEL 1: CONCEPTS

Q1. State the universal law of gravitation.

Q2. What is the difference between mass and weight?

Q3. Why does an object weigh less on the Moon than on the Earth?

Q4. What is buoyant force (upthrust)?

Q5. State Archimedes' principle.

LEVEL 2: NUMERICALS

Q6. An object has a mass of 10 kg. What is its weight on the Earth, and roughly what would it weigh on the Moon?

Q7. A ball is thrown vertically upwards with a speed of 49 m/s. Find (a) the maximum height it reaches and (b) the total time before it returns to the thrower.

Q8. A stone is released from the top of a tower 19.6 m high. Find its speed just before it hits the ground.

Q9. The relative density of silver is 10.8, and the density of water is 1000 kg/m^3. Find the density of silver.

Q10. A sealed packet has a mass of 500 g and a volume of 350 cm^3. Will it float or sink in water (density 1 g/cm^3)? What mass of water will it displace?

LEVEL 3: CHALLENGE (FOUNDATION)

Q11. How does the gravitational force between two objects change if (a) the mass of one is doubled, (b) the distance between them is doubled, (c) both masses are doubled?

Q12. A stone is dropped from a cliff and hits the ground after 3 s. Find the height of the cliff and the speed of the stone on impact.

Q13. Why does a block of plastic released under water come up to the surface, while an iron nail sinks?

~ Mass never changes. Weight does! ~